# Tyler Kuipers

Software Architect

*Edmonton, Canada*

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## About Me

In 2024, my team won the Guidewire Innovation Award for Canada's first pay-as-you-go auto insurance product, one of three worldwide. I've been writing software for about 10 years, with the last three focused on backend architecture, distributed systems, and platform infrastructure. I own the technical architecture for a platform managing $300M ARR: 4 production REST APIs, a self-service developer platform I built solo (60x provisioning improvement, adopted by 3 teams), and multiple AWS-native services on Kubernetes running at 99.9% uptime. I take these things from design through delivery and stay responsible for them. Technical leadership runs alongside the IC work: I'm on the Architecture Steering Committee, I've authored RFCs on development practices and system architecture, and I set technical direction for a 12-person team through code reviews, PR standards, and design guidelines. The work I find most satisfying is the kind that makes other engineers faster or lowers the cost of getting things wrong.

## Work Experience

### Alberta Motor Association

*Edmonton, Alberta*

Software Developer IV

*Jan 2023 - Present*

- Architected the infrastructure for Canada's first pay-as-you-go insurance product across multiple AWS accounts, owned delivery from design to production, and kept it running: 10 Lambda functions, 99.9% uptime since launch, 2.8M insured trips and 76,637 invoices processed, $4.3M annual revenue. Won the 2024 Guidewire Innovation Award (1 of 3 worldwide). (Python, TypeScript, AWS Lambda, Terraform)
- Built a self-service developer platform solo that took environment provisioning from 2-3 weeks down to 2 hours (60x). Adopted by 15 people across 3 teams, used to spin up 120+ environments. Enabled production debugging that was previously impossible: a cancellation bug nobody could reproduce got root-caused in days using snapshotted production data. (Python, Terraform, AWS, Jenkins, Docker)
- Member of the Architecture Steering Committee: a quarterly cross-org body that reviews and approves architectural proposals, with advisory and decision-making authority over architectural direction.
- Authored an RFC to shift the team to trunk-based development with feature flags: `develop` as single source of truth, features deployed dormant and toggled active. Decouples deployment from release, eliminates code reversions, enables instant rollback. Currently in pilot.
- Own 4 REST APIs serving 5 internal teams and external partners. Designed the contracts and versioning, approve new endpoint additions, and maintain OpenAPI design standards in Confluence. Set coding standards for the 12-person team through code reviews, PR checklists, and API design guides. (Python, TypeScript, REST, OpenAPI)
- Led a 2-3 engineer rewrite of a brittle legacy C# document generation system into containerized ASP.NET Core microservices (.NET 8). 4-5 services on a managed Kubernetes cluster, deployed with Helm. Instrumented with Datadog metrics for failure rate, throughput, generation time, and storage time. (C#, ASP.NET Core, Kubernetes, Docker, Helm, Datadog)
- Built an event-driven pipeline that ingests an external partner's policy data: S3 ingestion, a Kubernetes CronJob, a translation service, and transactional calls to a composite API, with OAuth 2.0 auth and Datadog monitoring. (Python, Kubernetes, AWS S3, OAuth 2.0)

### Arterys Inc. (Acquired by Tempus Labs)

*Calgary, Alberta (Remote)*

Software Developer

*Aug 2021 - Aug 2022*

- Built and maintained backend services that ran medical AI algorithms on clinical imaging data and returned results to the platform. (TypeScript, Node.js, AWS)
- Improved AI algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, catching and resolving issues before customer impact.

### Alberta Motor Association

*Edmonton, Alberta*

Software Developer | Software Team Lead

*Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021*

- Designed and built automated test framework that reduced defects by 30% on subsequent projects. Adopted across the insurance engineering team ( 15 developers / all Guidewire projects) and still in use 4+ years later. (Gosu/Java)
- Led 5-6 person development team on Guidewire insurance platform using TDD methodology. Performed Scrum Master role. Delivered projects on time and budget with no critical defects post-launch. Code reviews and mentoring junior developers.
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local Guidewire development environments. Trained 5-6 developers on TDD methodologies through hosted training sessions.

### Pleasant Solutions

*Edmonton, Alberta*

Software Developer

*Jan 2018 - Jun 2018*

- Built HMI for a concrete plant control system using TDD methodology. Daily work in C# / .NET, heavily utilizing the MVC design pattern, with REST integration and tests written with NUnit and Moq. (C#, .NET, MVC, REST)

**Lockheed Martin Canada** — *Halifax, Nova Scotia*
Associate Software Engineer — *Jan 2017 - Jan 2018*

- Contributed to Data Fusion module for Combat Management Systems, combining data from multiple underwater and above-water sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system that enabled re-running failed test scenarios at will, improving debugging and testing capabilities.

**The King's University** — *Edmonton, Alberta*
Software Developer/Research Assistant — *May 2014 - Sep 2016*

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.
- Presented research through poster presentations.

## Education

**The Open University** — *Milton Keynes, England (Remote)*
MSc. Computing (Software Engineering) — *May 2024 - Ongoing*
Currently pursuing Master's in Computing at The Open University.

**The King's University** — *Edmonton, Alberta*
B.S. in Computer Science — *Sep 2012 - Dec 2016*

## Skills

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| **Technical Leadership** | I shape technical direction through RFC authorship, Architecture Steering Committee membership, and cross-team influence rather than through direct reports. I've authored RFCs on development practices and system architecture, defined API contracts and coding standards for a 12-person team, and set up systems that outlast me: a test framework I designed in 2021 is still in use across the insurance engineering team four years later. |
| **Distributed Systems & Infrastructure** | I've spent the last three years designing and building distributed systems on AWS: multi-account architectures, event-driven pipelines, microservice deployments on Kubernetes/EKS, and service meshes with Linkerd. A self-service developer platform I built solo cut environment provisioning from 2-3 weeks to 2 hours (60x), adopted across 3 teams and used to spin up 120+ environments. |
| **Reliability & Observability** | I instrument what I ship with Datadog metrics for failure rate, throughput, and latency, and use them to set clear bars for what healthy looks like. The pay-as-you-go product I architected has run at 99.9% uptime since mid-2024 launch. I like chasing hard production failures to root cause: a cancellation bug nobody could reproduce got fixed in days once I could replay real production data against it. |
| **Software Craftsmanship** | TDD has been part of how I work since 2018. I designed a test framework that reduced defects by 30% on subsequent projects, and I care about CI/CD, writing code that's easy to change, and understanding root cause when something breaks. When a bug gets through, I want to know why and close the gap. |

## Technologies

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| **Backend Languages** | TypeScript, Node.js, Python, Java / Gosu (JVM), C# / .NET (ASP.NET Core, .NET 8), Bash |
| **Architecture & APIs** | RESTful APIs, OpenAPI, Microservices, Event-Driven Architecture, API Versioning |
| **Cloud & Infrastructure** | AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM, ECR), Terraform, Kubernetes, Docker, Azure DevOps, Azure Artifacts |
| **Databases** | PostgreSQL, SQL, Aurora/RDS, Data Modelling, Redis |
| **AI-Assisted Development** | Claude Code, Cursor, GitHub Copilot, Agentic Workflows |
| **Observability** | Datadog, Elasticsearch, Kibana, Metrics Design, SLOs, Alerting |
| **CI/CD & Automation** | CI/CD, GitHub Actions, Jenkins, GitOps, Feature Flags, Infrastructure as Code |
| **Testing** | NUnit, Moq, JUnit, TDD, Test Framework Design |